Investigation of the effects of indentation depth, gel thickness, and hyperelastic properties on the estimation of elastic modulus for a gel

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Solid Biomechanics Coursework

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1. Abstract

This study investigates the mechanical behaviour of soft gel-like biomaterials under micro-indentation using finite element simulations in ANSYS. The objective was to evaluate how indentation depth, gel thickness, and constitutive material models affect the estimation of Young's modulus. A Mooney-Rivlin hyper elastic model was implemented to capture nonlinear elastic behaviour, and results were compared against a linear elastic model with the same initial Young's modulus. Simulations revealed that gel thickness significantly influences modulus estimation due to substrate effects, with thinner gels leading to large overestimation unless indentation depth remains below 10% of the sample height. Furthermore, comparison between hyper elastic and linear elastic models demonstrated that the latter fails to represent large-strain behaviour accurately, therefore underestimating force and strain responses. These findings emphasize the importance of choosing suitable constitutive models and geometric constraints when characterising soft biological materials.

2. Background & Literature Review

2.1 Micro-Indentation as a Biomechanical Testing Technique

Cellular micro-indentation is an experimental technique used to quantify mechanical properties of soft materials, including gels and biological tissues. Such properties, including the Young's Modulus or effective viscosity, aid in understanding physiological and pathological cellular processes. The technique consists of applying a force on a sample using an indenter of specific geometry and analysing the resulting indentation [1].

Conventional experimental approaches are used to characterise materials. On a macroscale, compression and tensile testing methods are conducted. On a nanoscale, atomic force microscopy (AFM) is used. It is a microrheology tool used to investigate surface structures, quantify interaction forces between them, and map the distribution of elastic properties [3]. AFM uses a cantilever with a sharp tip that deforms the sample, allowing force-displacement curves to be recorded [2]. Often, Hertz contact theory – i.e. an analysis of the advancing contact of two deformable solids with smooth shapes and no friction – is used to obtain the elastic modulus from the indentation load and indentation depth curves [5].

Compared to conventional methods, micro-indentation has several advantages. Sample preparation requires less effort: consistent dimensions and uniform cross-sectional areas between samples are not needed, in turn increasing testing accuracy and limiting damage to soft tissue samples, allowing them to be tested in-vivo [6]. However micro-indentation has limitations, particularly because it relies on mathematical indentation models to interpret mechanical properties from the obtained data, which are challenging to derive due to the variety of indenter shapes (spherical, pyramidal, conical). Accuracy is also affected by the large number of assumptions (negligible friction or incompressibility). For instance, Hertzian models assume infinite sample thickness, which is unrealistic for thin gels [6, 7].

Several studies have refined micro-indentation models to improve elastic modulus estimation. Dimitriadis et. Al. (2002) extended Hertz contact theory by incorporating substrate effects, specifically for thin layers attached to rigid supports, a crucial aspect when analysing soft biomaterials on glass substrates [7]. This correction provided measurements with better accuracy by addressing deviations caused by finite thickness and indentation depth.

2.2 Hyper-elasticity in Soft Biological Materials

Hyper elasticity describes the behaviour of materials such as elastomers, rubbers and biological tissues. They undergo large deformations (100 to 700%) which are reversible upon unloading, all while maintaining an elastic response. They exhibit highly nonlinear stress-strain relationships with a varying stiffness as they deform, unlike linear elastic materials which follow Hooke's Law and have a constant stiffness [3]. Hyper elastic materials are usually nearly incompressible: their volume does not significantly change under deformation. Their mechanical response differs in tension and compression: they are initially soft and then stiffen under tensile load, whereas they remain stiff under compressive forces [3].

Several hyper-elastic material models exist, including Mooney-Rivlin, Neo-Hookean, Ogden, Yeoh, Green, and Simpson models. These are used to predict the non-linear behaviour of materials based on the strain energy density function, which describes the material's response to deformation in terms of stored elastic energy [8]. These models have been applied in biomedical and tissue engineering: for instance, a study by Yazdi et al. shows their application in the mechanical modelling and characterisation of human skin, which exhibits hyper-elastic behaviour [8]. Such studies enable research, diagnosing and monitoring of skin conditions.

2.3 Finite Element Modelling of Indentation Tests

Finite Element tools on ANSYS can be used to investigate effects of gel thickness, indentation depth, and hyper-elastic properties for estimating elastic moduli. A study by Ayyalasomayajula et al. shows that an inverse finite element analysis is required to identify the parameters of the hyper elastic material properties due to the lack of information in the indentation test alone [4].

2.4 Influence of Gel Thickness and Indentation Depth on Modulus Estimation

Substrate effects significantly influence modulus estimation in indentation tests on thin gel layers. When indentation depth is not small compared to gel thickness, the rigid substrate increases the measured stiffness, leading to overestimation of Young's modulus. Dimitriadis et al. showed that the Hertz model [7], which assumes infinite thickness, is inaccurate for such cases, and proposed correction factors for finite-thickness layers, both bonded and unbonded. They also recommended keeping indentation depth below 10% of the sample thickness to reduce substrate influence and ensure accurate modulus extraction in soft, thin biological samples.

3. Aims and Objectives

The aims and objectives of this project are:

- To conduct a computer simulation of micro-indentation using a Finite Element Analysis (FEA) methodology, on the sophisticated software ANSYS.
- Conduct a literature review and research key parameters, including the gel, glass and indenter materials, geometries, thicknesses, indentation depth ranges.
- Create a testing methodology for understanding the variation of Young's modulus for different indentation depths, gel thicknesses and hyper-elastic properties. This includes ensuring correct mesh sizing, boundary conditions and connections, obtaining stress-strain curves, and any other relevant data for analysis.
- Provide plots, tables and images where necessary for visualisation and better explainability.

4. Methodology

The following section describes the methodology used to conduct the study, designed using ANSYS. To begin, a static structural analysis system was chosen, in which three components are defined: the rigid substrate, the hyper-elastic gel, and the indenter. All parameters were chosen based on current literature and previous studies.

Engineering Data

The first parameters which needed to be defined in this study were the materials for every component. Common materials for the indenter top are monocrystalline silicon and silicon nitride. However occasionally**,** tungsten, nickel or other strong materials can be used. Therefore, stainless steel was chosen to conduct the experiment. For the rigid substrate, it was found that glass or mica can be used, therefore glass was chosen [7, 10]. Finally, to characterise the gel, the Mooney-Rivlin hyper-elastic model was found to provide the best accuracy for hyper-elastic gels [11]. Using a built-in model in ANSYS, two material constants C_{10} and C_{01} needed to be determined, as well as an incompressible parameter D_1. Assuming small deformations and incompressibility, the shear modulus G is:

$$G = 2(C_{10} + C_{01})$$

Where $C_{10} + C_{01}$ are the material constants of the Mooney-Rivlin equation. Knowing the Young's modulus E can be expressed as:

$$E = 2G(1 + \nu)$$

Assume an incompressible material so $\nu \approx 0.49$, then:

$$E \approx 3G$$
$$\frac{E}{3} \approx G$$
$$\frac{E}{3} \approx 2(C_{10} + C_{01})$$

Assuming $C_{10} = C_{01}$ for simplicity, it can be approximated to:

$$\frac{E}{3} \approx 4C_{10}$$
$$C_{10} \approx \frac{E}{12}$$

The values 1, 5 and 20 kPa were chosen respectively for the Young's modulus, as they are typical values for biological tissues such as cells [7].

For the incompressibility parameter D_1, we know the bulk modulus can be expressed:

$$K = \frac{E}{3(1-2\nu)}$$

And it can be found that [12]:

$$D_1 = \frac{2}{K}$$

Numerical values for these parameters are shown in *Fig. 1*.

Geometry

Three 2D-shapes were drawn on the XY Plane, and the behaviour type was set to axisymmetric. The rigid glass substrate and glass were each represented by a rectangle of length $100\ \mu\text{m}$ and height $1\ \mu\text{m}$. The two rectangles were stacked on top of each other. Examples of indenter geometries currently used in literature include sharp or pyramidal tips, however they have shown damage to the contact region [7]. Spherical indenter geometry was therefore chosen, as it is efficient for small strains and indentations [7]. For this, a semi-circle of radius $5\ \mu\text{m}$ [7] was drawn in the middle of the rectangles, such that the centre of the circle is at an equal distance of $50\ \mu\text{m}$ from both edges.

Model

Effectively setting up the model was then needed to run simulations. The defined geometries were imported, then two manual contact regions were fixed. The first was "frictionless" between the indenter (contact body) and gel (target body), and the second was "bonded" between glass (contact body) and gel (target body). Boundary conditions were then defined such that:

- Glass substrate is a fixed support
- Left edge, right edge, and top of the gel are fixed with no displacement on the x-axis ($X = 0\ m$), but the y-axis had free displacement ($Y = free$)

Meshes with element sizes of $1\ \mu\text{m}$ were then set up for the indenter and the glass substrate, and element sizes of $0.1\ \mu\text{m}$ for the gel. Quadrilateral dominant meshing was added for higher accuracy per element. This is shown in Fig. 3.

Testing

Three experiments were conducted to assess the impact of indentation depth, gel thickness and hyper elastic properties on E.

1) Different indentation depths, ranging from $0.1\ \mu\text{m}$ to $0.7\ \mu\text{m}$. Values were chosen on the basis that indentation depth should be less than 10% of the sample height [13].
2) Different gel thicknesses, which should be between $1\ \mu\text{m}$ and $4\ \mu\text{m}$ for thin gels [7].
3) The effects of a hyper-elastic material versus a linear elastic material for the gel. Properties are detailed on Fig. 2.

5. Results and Discussion

Having recorded the force and displacement, the Hertz contact theory can be used to find the Young's Modulus E such that:

$$F = \frac{4}{3}\frac{E}{(1-v^2)}\sqrt{R}\delta^{\frac{3}{2}}$$

Where F is the force found using the finite element simulations, δ is the indentation depth, v is the Poisson ratio, and R is the indenter radius.

For test 1, the force-indentation curve was first plotted. This is shown in *Fig. 4*. The trend obtained experimentally matches that of the theory. Therefore, the second step was to calculate the Young's modulus, and plot it in terms of indentation depth. Isolate E:

$$E = \frac{3F(1-v^2)}{4\sqrt{R}\delta^{\frac{3}{2}}}$$

Fig. 5 (left plot) shows that the experimentally obtained E displays a non-monotonic trend with indentation depth: it starts by decreasing until it reaches a minimum at around $0.2 - 0.25\ \mu\text{m}$, then increases and plateaus at deeper indentation levels. While viscoelastic materials typically exhibit a decrease in modulus due to a time-dependent relaxation, this increase at larger depths highlights the possibility of other factors affecting results.

One likely factor could be the use of the Mooney-Rivlin model, which captures nonlinear elasticity but can become unstable under large compressive strains (when going above 10%) or when the parameter C_{01} is not defined properly (in this case, it could be because of the assumption that $C_{10} = C_{01}$). Literature shows [14] that while these parameters fit tensile data well, they can cause non-physical behaviour in compression (right plot of *Fig.5*). Since indentation involves compressive loading, this could explain the observed dip. The initial decrease is due to rapid relaxation near the surface, which is what we expect from theory, and the subsequent increase is possibly due to the engagement of bulk material and strain-stiffening properties, due to improper modelling.

For test 2, for a fixed indentation depth of $0.2\ \mu\mathrm{m}$, the same formula can be used to calculate Young's modulus. The plot in *Fig. 6* was obtained, for which we can see that as gel thickness increased from 0.4 to $0.6\ \mu\mathrm{m}$, the apparent Young's modulus decreased sharply. This indicates a significant substrate effect in thinner gels. Beyond approximately $2\ \mu\mathrm{m}$ of gel thickness, E values stabilise near the expected theoretical value of 10 kPa, with overestimation falling below 10% (19276.86% overestimation at $0.8\ \mu\mathrm{m}$ vs 2.86% overestimation at $2\ \mu\mathrm{m}$). This aligns with the established guideline that the indentation depth should be less than 10% of the sample thickness to avoid substrate influence during AFM-based mechanical measurements. Fig. 7 shows an example of stress and strain for a fixed indentation depth.

For test 3, a hyper elastic (Mooney-Rivlin) model was compared with a linear elastic model. Major differences were seen when looking at gel behaviour under large indentation. With the same initial Young's modulus, the hyper elastic model produced much higher stress, strain and force values as well as a localised deformation under the indenter, as seen on Fig. 8. This characterises real soft gels in contrast to the linear elastic model which assumes a constant stiffness, underestimated deformation and unrealistically distributed strain. The results therefore confirm that linear elasticity cannot accurately represent soft gel mechanics at large strains, and that hyper elastic models are necessary to capture non-linear stiffening and realistic force-deformation responses.

6. Conclusion

In conclusion, FEA-based simulations of micro-indentation highlighted key factors affecting the mechanical characterisation of soft gels. It was found that indentation into thin gels leads to overestimation of Young's modulus due to the influence of the rigid substrate. This effect diminishes when indentation depth remains below 10% of the gel thickness. Additionally, the use of a hyper elastic Mooney-Rivlin model captured nonlinear stiffening and localised deformation typical of soft gels under large strain. In contrast the linear elastic model strongly underestimated both stress and strain, confirming its inadequacy for large-deformation scenarios. These insights validate the importance of using hyper elastic constitutive models and appropriate geometrical considerations in the mechanical modelling of soft biological tissues.

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Appendix A: Material definition

Properties of Outline Row 7: Structural Steel

	A	B	C
1	Property	Value	Unit
2	Material Field Variables	Table	
3	Density	7850	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Young's Modulus a...	
8	Young's Modulus	2E+11	Pa
9	Poisson's Ratio	0.3	
10	Bulk Modulus	1.6667E+11	Pa
11	Shear Modulus	7.6923E+10	Pa
12	Strain-Life Parameters		
20	S-N Curve	Tabular	
24	Tensile Yield Strength	2.5E+08	Pa
25	Compressive Yield Strength	2.5E+08	Pa
26	Tensile Ultimate Strength	4.6E+08	Pa
27	Compressive Ultimate Strength	0	Pa

Properties of Outline Row 4: Glass substrate

	A	B	C
1	Property	Value	Unit
2	Material Field Variables	Table	
3	Isotropic Elasticity		
4	Derive from	Young's Modulus...	
5	Young's Modulus	70	GPa
6	Poisson's Ratio	0.2	
7	Bulk Modulus	3.8889E+10	Pa
8	Shear Modulus	2.9167E+10	Pa

Properties of Outline Row 3: Gel material

	A	B	C
1	Property	Value	Unit
2	Mooney-Rivlin 2 Parameter		
3	Material Constant C10	1666.7	Pa
4	Material Constant C01	1666.7	Pa
5	Incompressibility Parameter D1	6E-06	Pa^-1

Figure 1. Structural steel properties used for AFM indenter tip on the top left. Glass properties used for rigid substrate on the top right, common values for glass Young's modulus (70 GPa) [10] and Poisson Ratio (0.2) [12] can be found. At the bottom are the gel properties, the C_{10} and C_{01} values (1666.7 Pa) were obtained for $E = 20\ kPa$.

Properties of Outline Row 6: Linear elastic

	A	B	C
1	Property	Value	Unit
2	Material Field Variables	Table	
3	Isotropic Elasticity		
4	Derive from	Young's Modulus and Poisson...	
5	Young's Modulus	20000	Pa
6	Poisson's Ratio	0.49	
7	Bulk Modulus	3.3333E+05	Pa
8	Shear Modulus	6711.4	Pa

Figure 2. Linear elastic property for testing hyper-elastic vs linear elastic gel materials. The values were chosen based on the previously explained assumptions.

Appendix B: Schematic and Meshes

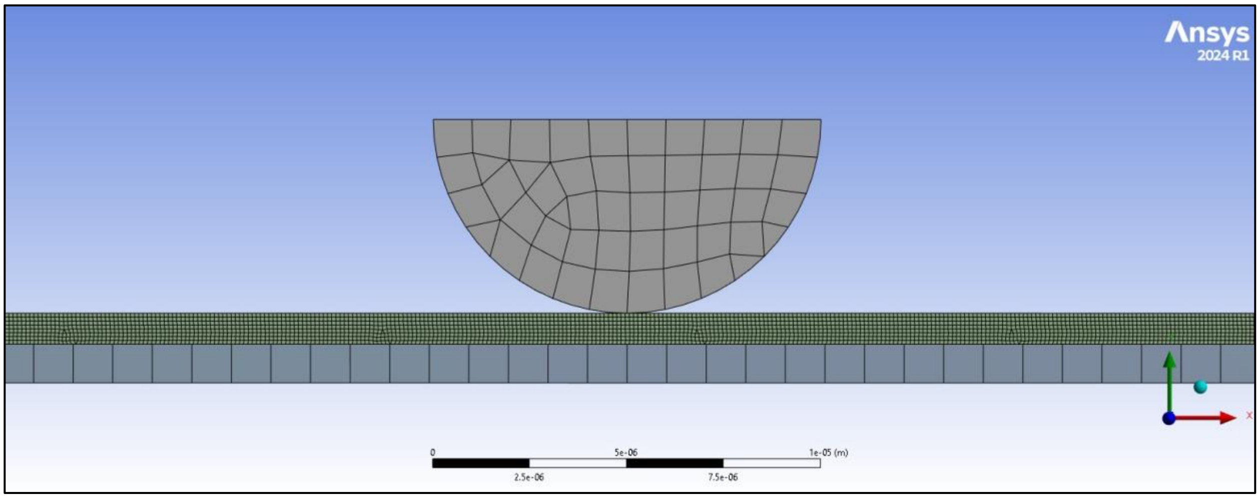


Figure 3. Meshes of $1\ \mu m$ for the indenter tip and glass substrate, and $0.1\ \mu m$ for the gel.

Appendix C: Results plots

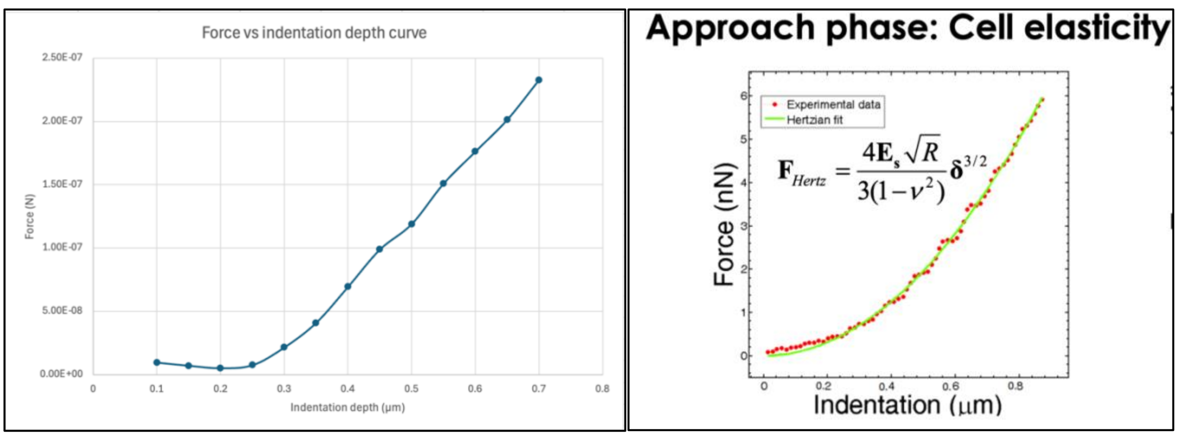


Figure 4. Force-indentation curve obtained experimentally (left) vs theoretical expectation from lecture 8 (right).

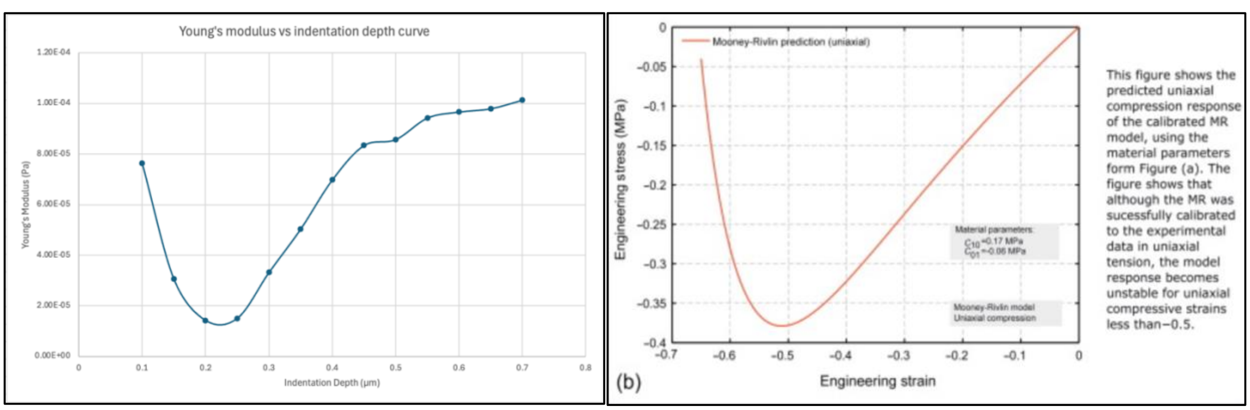


Fig. 5 shows the trend obtained experimentally when rearranging the Hertz equation for the Young's modulus (left). This behavior was compared to an existing study (right) found in Chapter 5 of Jörgen Bergström's book, *Mechanics of Solid Polymers* [14].

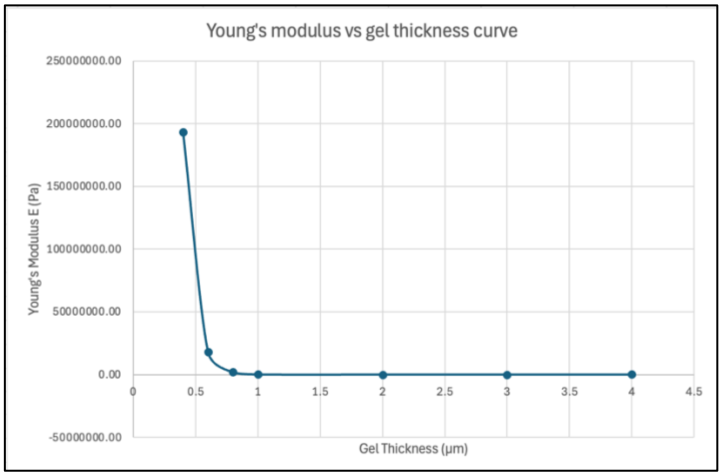


Figure 6. Young's modulus vs gel thickness curve.

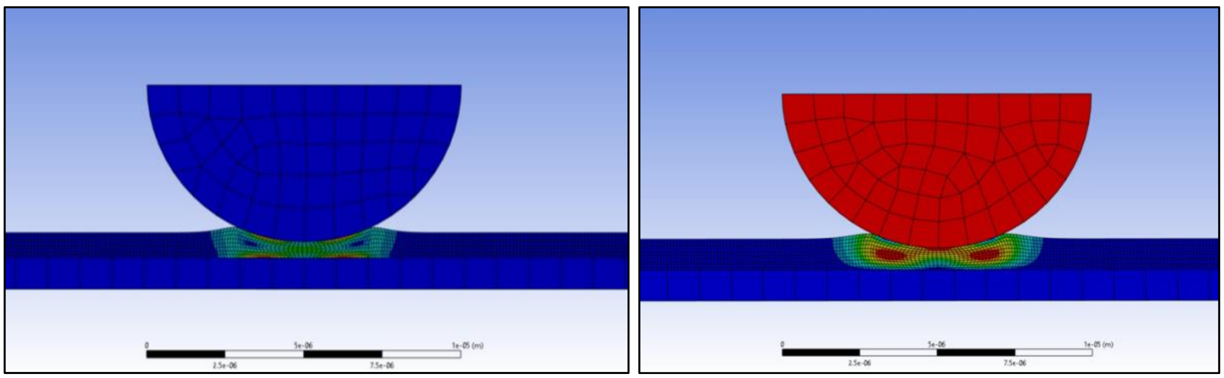
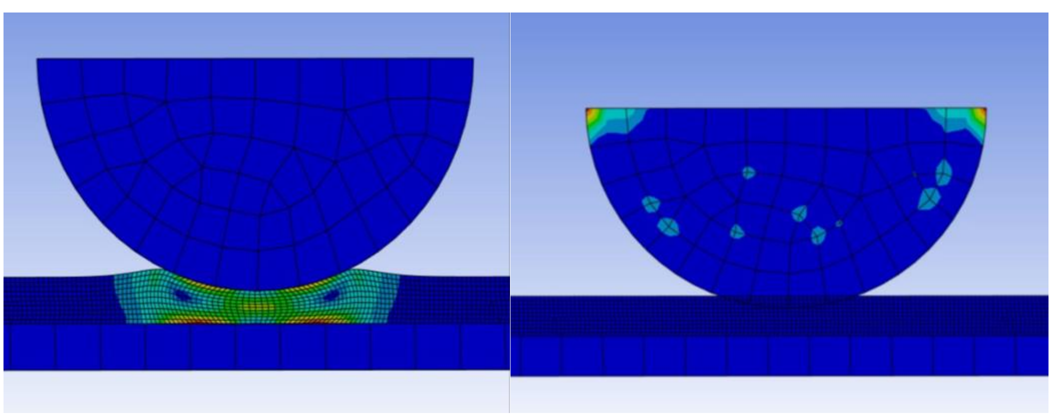


Figure 7. Example of stress (left) and total deformation (right)



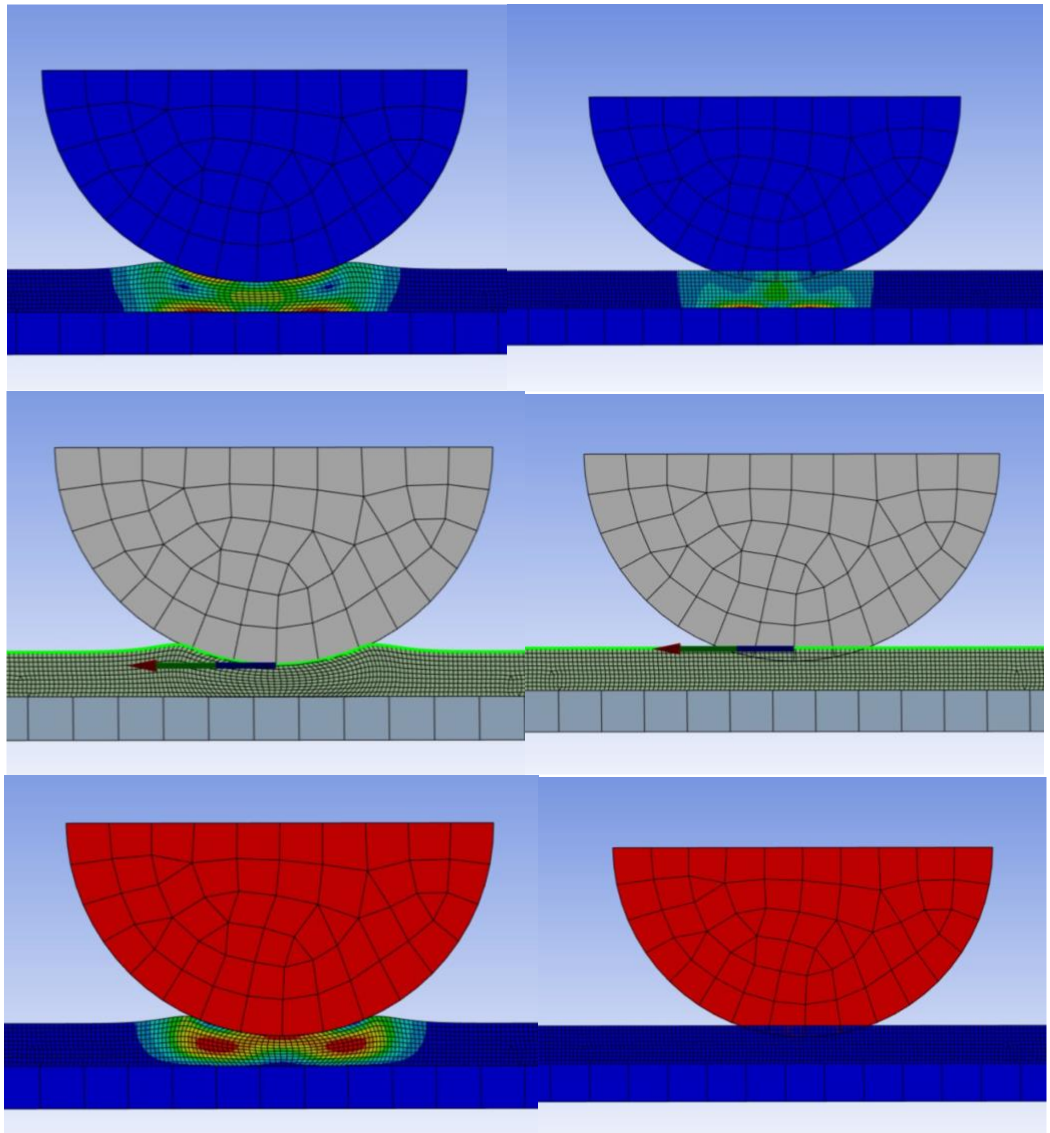


Figure 8. From top to bottom: stress (first), strain (second), reaction force (third), and localised deformation (last). Comparison between hyper elastic gel material (left) vs linear elastic gel material (right).

Appendix D: ANSYS Measurements

Experiment 1: Indentation Depths

For $E = 10\ kPa$, $C_{10} = C_{01} = 833.33\ Pa$ and $D_1 = 1.2 \times 10^{-5}\ Pa^{-1}$, and for a gel thickness of 2 μm:

Indentation Depth	Stress (Pa)	Strain (m/m)	Total Deformation (m)	Max. Force (N)
$0.1\ \mu m$	412.85	4.3321e-2	8.7544e-8	9.4632e-9
$0.15\ \mu m$	642.14	6.6926e-2	1.3193e-7	6.9696e-9
$0.2\ \mu m$	883.19	9.1442e-2	1.7557e-7	5.0025e-9
$0.25\ \mu m$	1184.7	0.12163	2.2065e-7	7.4027e-9
$0.3\ \mu m$	1522	0.1546	2.6504e-7	2.1412e-8

0.35 μm	1889	0.18947	3.0898e-7	4.0894e-8
0.4 μm	2259.8	0.22325	3.526e-7	6.9296e-8
0.45 μm	2632.6	0.25581	3.9658e-7	9.8773e-8
0.5 μm	3018.5	0.2881	4.4123e-7	1.1893e-7
0.55 μm	3406.5	0.31927	4.8238e-7	1.5085e-7
0.6 μm	4200.4	0.38024	5.5246e-7	1.0973e-7
0.65 μm	4902.8	0.43029	6.1212e-7	2.0136e-7
0.7 μm	5846.8	0.49183	6.4964e-7	2.3281e-7

Experiment 2: Gel thicknesses

For $E = 10\ kPa$, $C_{10} = C_{01} = 833.33\ Pa$ and $D_1 = 1.2 \times 10^{-5}\ Pa^{-1}$, and for an indentation depth of $0.2\ \mu m$.

Gel Thickness	Stress (Pa)	Strain (m/m)	Total Deformation (m)	Max. Force. (N)
0.4 μm	180580	1.3696	6.374e-7	6.7739e-5
0.6 μm	44292	1.5946	3.4214e-7	6.3727e-6
0.8 μm	23591	1.1471	5.2653e-7	6.7647e-7
1 μm	1434.3	0.15891	1.646e-7	4.3128e-8
2 μm	883.19	9.1442e-2	1.7557e-7	1.0025e-10
3 μm	720.94	7.4886e-2	1.6993e-7	7.2221e-9
4 μm	607.18	6.2253e-2	1.7815e-7	1.0062e-8
5 μm	533.23	5.5082e-2	1.8351e-7	1.6553e-8
6 μm	477.64	5.006e-2	1.8092e-7	3.458e-8

Experiment 3: Hyper elastic properties vs linear elastic properties

For a gel thickness of $1\ \mu m$, and an indentation depth of $0.3\ \mu m$ (30% indentation > 10%). Mesh of 0.1 micrometer for Gel.

Young's Modulus E	C10 (Pa)	C01 (Pa)	D1	Stress [Pa]	Strain [m/m]	Total deformation [m]	Max. Force [N]
20 kPa	1666.67	1666.67	6e-6	8804.7	0.67526	3.0147e-7	1.384e-7

Then, for an isotropic elastic property with a Poisson ratio of 0.49 and Young's modulus of $E = 20\ kPa$.

Young's Modulus E	Poisson	Stress [Pa]	Strain [m/m]	Total deformation [m]	Max. Force [N]
20 kPa	0.49	1.8673e-2	2.2671e-10	3.0e-7	9.5987e-16